

PAPER • OPEN ACCESS

Artificial intelligence system for the analysis of theoretical and experimental current-voltage characteristics

To cite this article: I V Gudza *et al* 2021 *J. Phys.: Conf. Ser.* **2131** 022089

View the [article online](#) for updates and enhancements.

You may also like

- [MaNGA DynPop. VII. A Unified Bulge–Disk–Halo Model for Explaining Diversity in Circular Velocity Curves of 6000 Spiral and Early-type Galaxies](#)
Kai Zhu, Michele Cappellari, Shude Mao et al.
- [Effects of assisted and variable mechanical ventilation on cardiorespiratory interactions in anesthetized pigs](#)
Alessandro Beda, Andreas Güldner, David M Simpson et al.
- [Transcranial Doppler of the middle cerebral artery indicates regional gray matter cerebral perfusion](#)
Evan P Pasha, Takashi Tarumi, Andreana P Haley et al.

Artificial intelligence system for the analysis of theoretical and experimental current-voltage characteristics

I V Gudza*, A V Kovalenko, M Kh Urtenov, and A V Pismenskiy

Kuban State University Krasnodar, 149 Stavropolskaya st. 350040 Russia

E-mail: flash.wetal@mail.ru

Abstract. Digitalization is one of the development priorities in the global scientific community. Digitalization at this point in time means the introduction of artificial intelligence systems in production, science, economics, management, etc. Artificial intelligence systems have shown their effectiveness in various fields of science and technology, including electrochemistry for such tasks as modeling membrane separation, controlling the variable operation of a simple seawater reverse osmosis plant, etc., however, the study and prediction of theoretical and experimental current-voltage characteristics of electromembrane systems by machine learning and artificial intelligence methods is still practically not carried out. The problem of finding connections between the regularities of changes in the current-voltage characteristics (CVC) and the regularities of the process of salt ion transfer in the electrodialysis apparatus (EDA) desalination channel, the classification of theoretical and experimental CVC depending on the values of the input parameters has not yet been set and investigated, and therefore is a new fundamental problem and an actual practical problem.

1. Introduction

Until now, the only way to study the CVC was to analyze the experimental data. The study of experimental fluctuations of the CVC in the works of Budnikov E. Yu., Kolganov V. I., Kolyubin A.V. by the methods of Fourier analysis [6], wavelet analysis [7], shows the complex, non-stationary and unstable behavior of the CVC.

For the theoretical study of this behavior of CVC, it is necessary to use special mathematical methods and models. We have made attempts to calculate the theoretical CVC and analyze them [8].

It should be noted that without the use of the latest mathematical methods and models, such as machine learning methods and artificial intelligence systems, it becomes more and more problematic to move forward in the development of the theory and application of membrane systems, since models are becoming more and more complex and become so complex that the compilation of algorithms and numerical solutions, engineering calculations of membrane systems are made disproportionately time-consuming, expensive, multiscale, as shown by V. Pham, Z. Li, J. White, J. Han and other foreign researchers [4, 5].

However, the study and prediction of theoretical and experimental CVC of membrane systems by machine learning and artificial intelligence methods has not yet been conducted.

2. Suggested approaches and methods

Previously, almost the only means in the arsenal of researchers of CVC and other characteristics, such as chronopotentiograms, chronogalvanograms and electrochemical noises, was the hardware Fourier



Content from this work may be used under the terms of the [Creative Commons Attribution 3.0 licence](https://creativecommons.org/licenses/by/3.0/). Any further distribution of this work must maintain attribution to the author(s) and the title of the work, journal citation and DOI.

transform, when a filter with a narrow bandwidth was connected to the system - source of electrical noise, which allowed measuring the noise power at a given frequency [6, 9], wavelet analysis [7, 10-13], the methods of dynamic chaos [14].

The result of these experiments is the power spectrum of the noise signal. The frequency range in which the measurements were made was 50 Hz-1 kHz.

CVC studies are constrained by the lack of adequate theoretical models suitable for interpretation of experimental data and understanding of the processes underlying CVC changes.

In this article, we have investigated theoretical and experimental CVC and created a CVC database containing more than 100 theoretical and 20 experimental CVC.

For the numerical calculation of the CVC, we used a modification of the basic mathematical model [15], which consists in the development and use of an algorithm for calculating the CVC.

This model is used to create training and control sets for a certain set of input parameters. To generate and analyze CVC for arbitrary input parameters, we have created neural networks and artificial intelligence systems in the Matlab and Statistica Neural Network environments. Experimental CVC was obtained for flow-through membrane cells under the conditions of the combined action of the main processes occurring in the EDA desalination channels.

We analyzed the obtained results, found a number of regularities, for example, the presence of several CVC sites with different behavior patterns, and explained the fluctuations of CVC, as well as verification and refinement of the obtained models.

The study was carried out at various values of input parameters, for example, the initial velocity of forced convection, the initial concentration, current density values, potential jump, etc.

We have determined the main patterns of salt ion transport in the EDA desalination channel, for example, the occurrence and development of the spatial charge, electroconvection, etc., using the created artificial intelligence system.

To implement the task of predicting new CVC based on input parameters using neural networks, various deep learning neural networks were created (multilayer perceptron, radial basis, convolutional, recurrent neural networks trained by gradient descent, Levenberg-Marquardt, etc. with Cross - Batch Statefulness [16].

For training neural networks "Deep Neural Network" (figure 1) databases of experimental and theoretical CVC were created (figure 2), containing all possible characteristic parameters of desalination channels of electromembrane systems (table 1) for theoretical and experimental CVC.

We analyzed more than 100 theoretical and 20 experimental current-voltage characteristics, and determined the most general patterns of various characteristic parameters of desalination channels of electromembrane systems (table 1). For example, we studied the dependence of the CVC on the channel length (figure 3), on the pumping rate of the solution (figure 4), on the presence and absence of spacers and their different shapes and sizes.

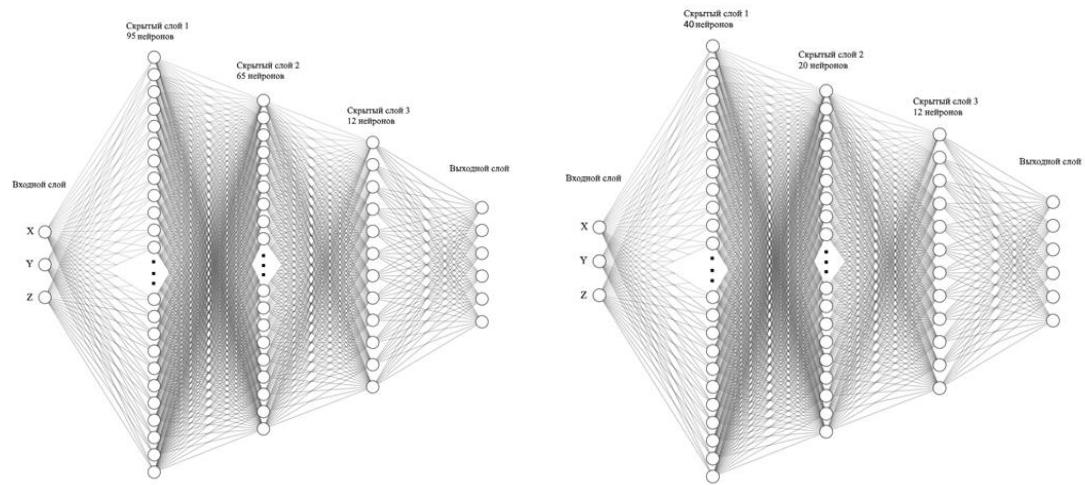


Figure 1. Neural networks "Deep Neural Network" of experimental and theoretical CVC.

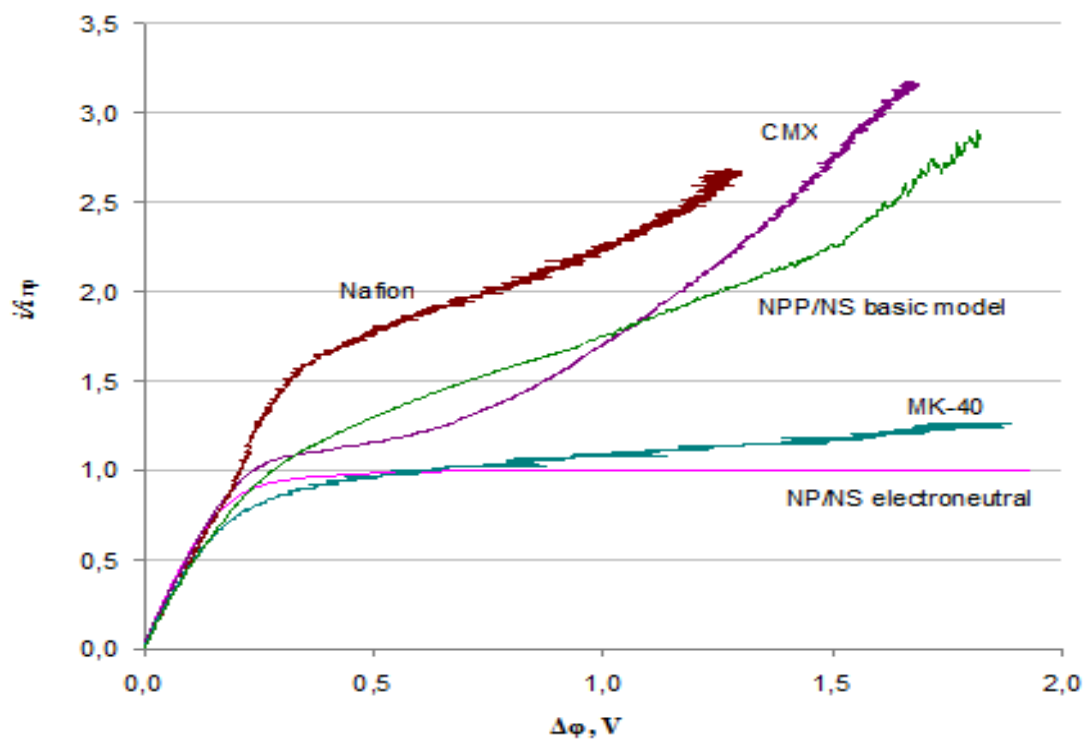


Figure 2. Comparison of calculated CVC (NP / NS electroneutral and NPP / NS basic model) and experimental CVC (Nafion, CMX and MK-40) [15].

Table 1. Variable parameters of desalination channels of electromembrane systems.

Parameter name	Parameter value [dimension]	Parameter description
H	$0.5 \div 1$ [mm]	cell width
L	$H \div 4H$ [mm]	cell length
V0	$8 \cdot 10^{-4} \div 10^{-2}$ [m/s]	initial speed
C0	$0.01 \div 1$ [mol/ m ³]	initial concentration
D1	$1.33 \cdot 10^{-9}$ [m ² /s]	diffusion coefficient cation
D2	$2.05 \cdot 10^{-9}$ [m ² /s]	diffusion coefficient anion
z1	1	charge number of cation
z2	-1	charge number of anion
dnu	$0.8937 \cdot 10^{-3}$ [Pa·s]	dynamic viscosity
dpr	$81 \cdot 8.85 \cdot 10^{-12}$ [F/m]	dielectric constant of electrolyte
R	8.3145 [J/(mol·K)]	universal gas constant
T	298.15 [K]	temperature
F	96485.33 [C/mol]	Faraday constant
rho	1002 [kg/ m ³]	fluid density
alfa	$0.001 \div 0.1$ [V/s]	potential jump growth rate

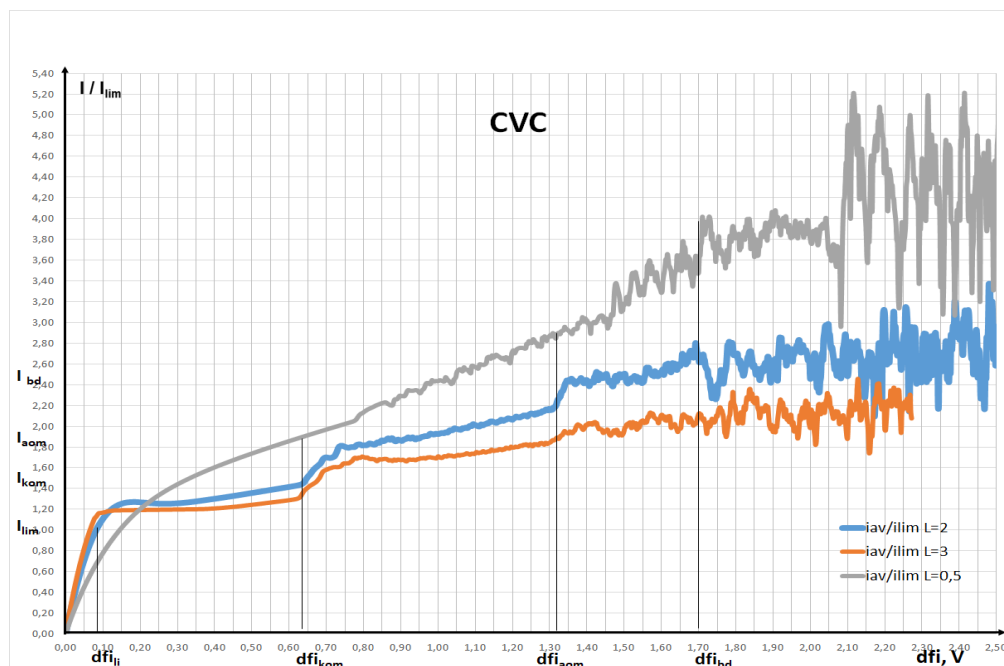


Figure 3. Current-voltage characteristics at $H = 0.5$, $V_0 = 0.1$ mm / s, $C_0 = 0.01$ mol / m³ and different channel lengths: "short" channel with $L = 0.5$ mm, middle channel $L = 2$ mm, "long channel" with $L = 3$ mm.

The following parameters were fed to the input of the "Deep Neural Network" neural networks (figure 1): x_1 – initial concentration, x_2 – solution pumping rate, x_3 – length and width of the EDA desalination channel. And at the output, we received critical values of the CVC such as: y_1 – a potential jump corresponding to the limiting current; y_2 – potential jump, corresponding to the beginning of electroconvection in CEM, y_3 – potential jump, corresponding to the beginning of electroconvection in AEM, y_4 – potential jump, corresponding to the beginning of the interaction of electroconvective vortices in AEM and CEM, y_5 – the critical value of the current corresponding to the beginning of electro-

convection in CEM(I_{CEM}), y_6 – the critical value of the current corresponding to the beginning of electroconvection in AEM(I_{AEM}). According to the obtained six indicators, the new CVC ("restored" CVC) was approximated and compared with the existing ones. The CVC approximation was based on the following principles. First, the ohmic section of the CVC corresponding to the prelimiting current (before the potential jump y_1) is a linear function with the corresponding slope angle. Secondly, the section of the CVC corresponding to the "plateau"(before the potential jump y_2) is also a linear function, but with a different angle of inclination. The remaining sections of the CVC (up to potential jumps y_2 , y_3 , y_4) are approximations of linear functions with the addition of fluctuations corresponding to random fluctuations in time series ("white" noise and "gray" noise). The created neural networks were trained by deep machine learning (Deep learning) in the Python development environment, using the OpenCV, TensorFlow, and Keras libraries.

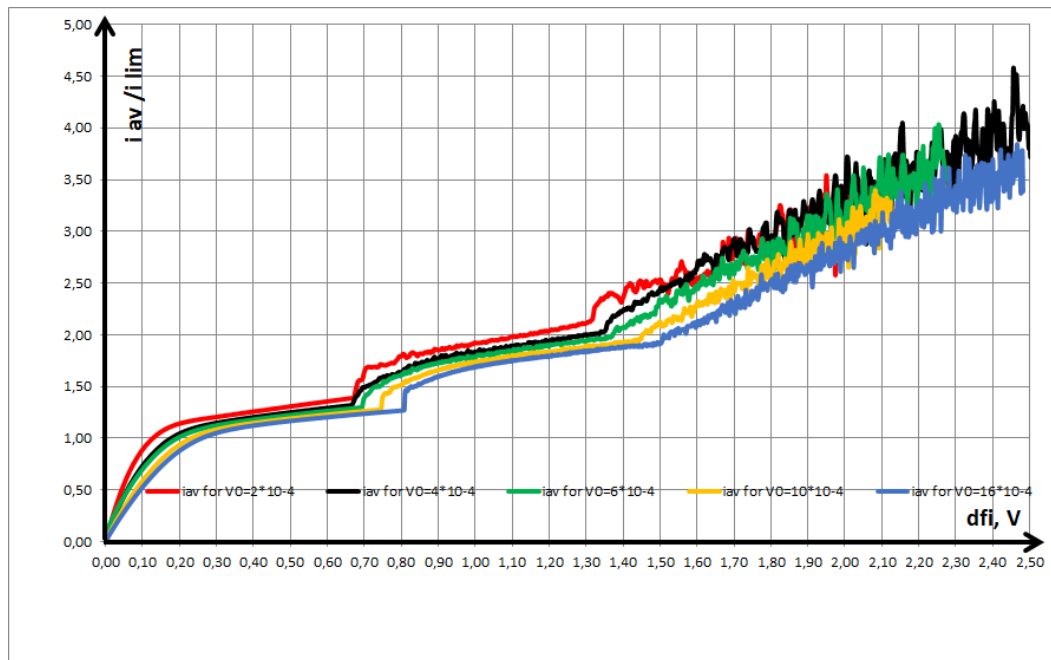


Figure 4. Current-voltage characteristics at $H = 0.5$, $C_0 = 0.01 \text{ mol / m}^3$ and different values of the speed of pumping the solution: $V_0 = 2 \cdot 10^{-4} \text{ m / s}$, $V_0 = 4 \cdot 10^{-4} \text{ m / s}$, $V_0 = 6 \cdot 10^{-4} \text{ m / s}$, $V_0 = 10^{-3} \text{ m / s}$, $V_0 = 1.6 \cdot 10^{-3} \text{ m / s}$.

The created artificial intelligence system for the analysis of theoretical and experimental current-voltage characteristics in electromembrane systems "AI CVC 0.1" allows, using the created neural networks, to obtain critical values and images of those CVC for which the calculated parameters are "unacceptable", i.e., too high concentration or the pumping speed of the solution, or the cell size. The calculation of such "unacceptable" conditions in the computer simulation environment of physical processes COMSOL Multiphysics 5.5 can take up huge computational resources.

The created artificial intelligence system for the analysis of theoretical and experimental current-voltage characteristics in electromembrane systems "AI CVC 0.1"(figure 5), allows us to obtain critical values of the CVC, to identify the trend and amplitude of vibrations of various parts of the CVC, to determine the main and accompanying frequencies of vibrations of various parts of the CVC, using the results obtained to record the "restored" CVC and compare it with the real one. The created artificial intelligence system for the analysis of theoretical and experimental current-voltage characteristics in electromembrane systems "AI CVC 0.1" allows you to calculate the critical values of currents (I_{CEM} ,

I_{AEM} , etc.), Hurst indicators for each section of the CVC, associated frequencies, oscillation amplitudes, tilt angles in the first and second sections.

The created neural networks "Deep Neural Network" of deep machine learning (Deep learning), allow to determine the critical values and recreate the CVC, using the approximation method described above. All created programs that make up the software package have certificates of state registration [17-19].

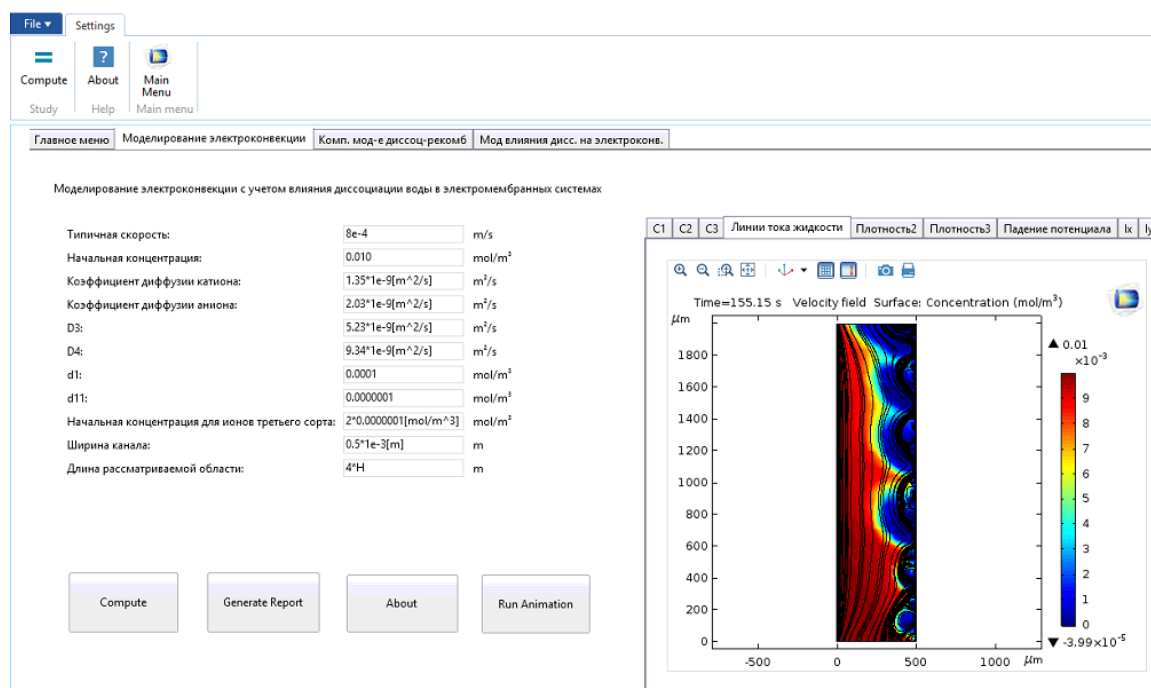


Figure 5. The created artificial intelligence system for the analysis of theoretical and experimental current-voltage characteristics in the "AI CVC 0.1" electromembrane systems with the "Electroconvection modeling" tab open".

Acknowledgments

The reported study was funded by RFBR according to the research project No. 19-08-00252 A.

References

- [1] Niemi H, Bulsari A, Palosaari S 1995 Simulation of membrane separation by neural networks *Journal of Membrane Science* **102** 185-191 DOI: 10.1016/0376-7388(94)00314-O
- [2] Cabrera P, Cartaa J A, González J, Melián G 2017 Artificial neural networks applied to manage the variable operation of a simple seawater reverse osmosis plant *Desalination* **416** 140-156 DOI: 10.1016/j.desal.2017.04.032
- [3] Kovalenko AV, Urtenov M 2020 Creation of an artificial intelligence system for analysis of theoretical Current-Voltage Curves *Trends Comput Sci Inf Technol* **5(1)** 005-007 DOI: 10.17352/tcsit.000011
- [4] Pham, Van et al. 2012 Direct Numerical Simulation of Electroconvective Instability and Hysteretic Current-voltage Response of a Permselective Membrane *Physical Review E* **86.4** DOI:10.1103/PhysRevE.86.046310
- [5] Kim B et al. 2016 Purification of High Salinity Brine by Multi-Stage Ion Concentration Polarization Desalination *Sci. Rep.* **6**, 31850 DOI: 10.1038/srep31850
- [6] Kolyubin A V, Maksimych A V, Timashev S F 1996 Using Flicker-noise spectroscopy to study the mechanism of the exorbitant current in a system with a cation exchange membrane *Elektrokhimiya* **32** 227

- [7] Budnikov E Yu, Kozlov S V, Kolyubin AV, Timashev S F 1999 Analysis of fluctuation phenomena in the process of electrochemical release of hydrogen on platinum *Journal of Physics. Chemistry* **73** 530
- [8] Kovalenko A V, Evdochenko E N, Urtenov M X 2015 Calculation and analysis of the time characteristics of electroconvection in membrane systems *Scientific Journal of KubGAU* **05(109)**
- [9] Kolganov V I, Malykhin M D, Babichev S V Akberova E M 2015 Analysis of the spectral composition of optical noise in a solution at the boundary with the MK-40 sulfocation exchange membrane after temperature exposure by Flicker-noise spectroscopy *VESTNIK VSU, SERIES: CHEMISTRY. BIOLOGY. PHARMACY* **3** 25
- [10] Budnikov E Yu Analysis of fluctuation phenomena in the field of exorbitant currents in an electromembrane system Diss. cand. phys. - math. n. m. 2000. 115 p.
- [11] Budnikov E Yu, Kukoev I Yu., Gulyaev A M, Miroshnikova I N, Maksimych A V, Timashev S F 1999 Wavelet and Fourier analysis of electrical fluctuations in semiconductor and electrochemical systems *Measuring Equipment* **11** 40
- [12] Budnikov E Yu, Maksimych A V, Kolyubin A V, Merkin V G, Timashev S F 1999 Wavelet analysis in the appendix to the study of the nature of the exorbitant current in an electrochemical system with a cation exchange membrane *Journal of Physical Chemistry* **73** 198
- [13] Egorov E V, Kozlov V A, Yashkin A V 2007 Phase-frequency characteristic of the transfer function of a spatially bounded electrochemical cell *Electrochemistry* **43(12)** 1
- [14] Asanov G S Dynamic chaos in nanostructured self-oscillating systems. Dissertation for the degree of Doctor of Philosophy (Ph. D.) in the field of radio engineering, electronics and telecommunications. Republic of Kazakhstan. Almaty, 2014. 108p
- [15] Urtenov M K, Uzdenova A M, Nikonenko V V, Pismenskaya N D, Kovalenko A V, Vasil'eva V I, Sistat P, Pourcelly G 2013 Basic mathematical model of overlimiting transfer enhanced by electroconvection in flow-through electrodialysis membrane cells *Journal of Membrane Science* **447** 190-202 DOI:10.1016/j.memsci.2013.07.033
- [16] Kovalenko A V, Urtenov M 2020 Creation of an artificial intelligence system for analysis of theoretical Current-Voltage Curves *Trends Comput Sci Inf Technol* **5(1)** 005-007 DOI: 10.17352/tcsit.000011
- [17] Kovalenko A V, Urtenov M A Kh, Sobchenko K V Software package for Fourier and wavelet analysis. Certificate of registration of the computer program RU 2017619076, 15.08.2017. Application no. 2017615871 dated 20.06.2017.
- [18] Kovalenko A V, Urtenov M A Kh, Kurzina A A, Seidova N M Calculation of the correlation exaltation of the limiting current of the generalized Sokirko-Harkats theory. Certificate of registration of the computer program RU 2018617186, 20.06.2018. Application no. 2018615458 dated 29.05.2018
- [19] Kovalenko A V, Urtenov M A Kh, Kurzina A A, Uzdenova A M A one-dimensional model for the numerical study of mass transfer processes taking into account the dissociation/recombination reaction of water molecules and Joule heating of a solution in membrane systems. Application no. 2018660257 dated 24.09.201